Fundamental Trigonometric Identities

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Using the Pythagorean identities**

Given that $\sin\theta = \frac{3}{5}$ and θ lies in Quadrant II, find:

a $\cos\theta = -\frac{4}{5}$ **b** $\tan\theta = -\frac{3}{4}$ **c** $\sec\theta = -\frac{5}{4}$ **d** $\csc\theta = \frac{5}{3}$

(a) $= -\frac{4}{5}$; (b) $= -\frac{3}{4}$; (c) $= -\frac{5}{4}$; (d) $= \frac{5}{3}$

2 **[Foundational]** • **Target: Using the Pythagorean identities**

Given that $\tan\theta = 2$ and θ lies in Quadrant III, find $\sin\theta$ and $\cos\theta$ in exact simplified form.

$\sec^2\theta = 1 + \tan^2\theta = 5$, and in QIII $\cos\theta < 0$, so $\cos\theta = -\frac{\sqrt{5}}{5}$ and $\sin\theta = \tan\theta\cos\theta = -\frac{2\sqrt{5}}{5}$.

3 **[Foundational]** • **Target: Simplifying and verifying**

Simplify each expression to a single trigonometric function:

a $\frac{1-\cos^2 x}{\sin x} = \frac{\sin^2 x}{\sin x} = \sin x$

b $\sec x\cot x = \frac{1}{\cos x} \cdot \frac{\cos x}{\sin x} = \csc x$

c $\tan x\csc x = \frac{\sin x}{\cos x} \cdot \frac{1}{\sin x} = \sec x$

(a) $= \frac{\sin^2 x}{\sin x} = \sin x$; (b) $= \frac{1}{\cos x} \cdot \frac{\cos x}{\sin x} = \csc x$; (c) $= \frac{\sin x}{\cos x} \cdot \frac{1}{\sin x} = \sec x$

4 **[Foundational]** • **Target: Simplifying and verifying**

Verify the identity $(1 + \tan^2 x)\cos^2 x = 1$.

$(1 + \tan^2 x)\cos^2 x = \sec^2 x\cos^2 x = \frac{1}{\cos^2 x}\cos^2 x = 1$.

5 **[Foundational]** • **Target: Quotient and reciprocal identities**

Rewrite each expression using only $\sin x$ and $\cos x$, then simplify:

a $\cot x\sin x = \frac{\cos x}{\sin x} \cdot \sin x = \cos x$

b $\frac{\sec x}{\csc x} = \frac{1/\cos x}{1/\sin x} = \frac{\sin x}{\cos x} = \tan x$

c $1 - (\sin^2 x\csc^2 x)\sin^2 x$ $1 - \sin^2 x = \cos^2 x$ (since $\csc^2 x\sin^2 x = 1$)

(a) $= \frac{\cos x}{\sin x} \cdot \sin x = \cos x$; (b) $= \frac{1/\cos x}{1/\sin x} = \frac{\sin x}{\cos x} = \tan x$; (c) $1 - \sin^2 x = \cos^2 x$ (since $\csc^2 x\sin^2 x = 1$)

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6 [Foundational] • Target: Quotient and reciprocal identities

Simplify $\frac{\cos x}{1-\sin x} - \tan x$ to a single term.

Combine over $\cos x(1-\sin x)$: $\frac{\cos^2 x - \sin x(1-\sin x)}{\cos x(1-\sin x)} = \frac{\cos^2 x + \sin^2 x - \sin x}{\cos x(1-\sin x)} = \frac{1-\sin x}{\cos x(1-\sin x)} = \sec x$.

7 [Foundational] • Target: More identity verifications

Verify the identity $\sec^2 x - \tan^2 x = 1$ using the Pythagorean identity for tangent and secant.

The identity $1 + \tan^2 x = \sec^2 x$ (dividing $\sin^2 x + \cos^2 x = 1$ by $\cos^2 x$) rearranges directly to $\sec^2 x - \tan^2 x = 1$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Identities with sums and products

Simplify $(\sec x - \tan x)(\sec x + \tan x)$ to a constant.

This is a difference of squares: $\sec^2 x - \tan^2 x = 1$ (a Pythagorean identity).

9 [Intermediate] • Target: Identities involving all three Pythagorean forms

State all three forms of the Pythagorean identity (for sine/cosine, tangent/secant, and cotangent/cosecant).

$\sin^2 x + \cos^2 x = 1$; $1 + \tan^2 x = \sec^2 x$; $1 + \cot^2 x = \csc^2 x$.

10 [Intermediate] • Target: Identities involving all three Pythagorean forms

Given $\cot\theta = \frac{5}{12}$ and θ in Quadrant I, find $\csc\theta$ and $\sin\theta$ using the cotangent-cosecant identity.

$\csc^2\theta = 1 + \frac{25}{144} = \frac{169}{144}$, and in QI $\csc\theta > 0$: $\csc\theta = \frac{13}{12}$. $\sin\theta = \frac{1}{\csc\theta} = \frac{12}{13}$.

11 [Intermediate] • Target: Simplifying more complex expressions

Simplify $\frac{\sin^3 x + \cos^3 x}{\sin x + \cos x}$ using the sum-of-cubes factoring pattern.

Factor as a sum of cubes: $\sin^3 x + \cos^3 x = (\sin x + \cos x)(\sin^2 x - \sin x\cos x + \cos^2 x)$. Dividing: $\frac{(\sin x + \cos x)(1 - \sin x\cos x)}{\sin x + \cos x} = 1 - \sin x\cos x$.

12 [Intermediate] • Target: Numeric identity verification (checking a specific angle)

Verify the identity $1 + \tan^2 x = \sec^2 x$ numerically at $x = \frac{\pi}{6}$ by computing both sides.

$\tan\frac{\pi}{6} = \frac{1}{\sqrt{3}}$: LHS $= 1 + \frac{1}{3} = \frac{4}{3}$. $\sec\frac{\pi}{6} = \frac{1}{\cos(\pi/6)} = \frac{2}{\sqrt{3}}$: RHS $= \frac{4}{3}$. Both sides match, confirming the identity at this angle (though this alone doesn't prove it for all x).

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13 [Intermediate] • Target: Identities with squared expressions

Given that $\sec\theta = -\frac{13}{5}$ and θ lies in Quadrant II, use the Pythagorean identity $1 + \tan^2\theta = \sec^2\theta$ to find $\tan\theta$ exactly, and then find $\sin\theta$ using the quotient identity.

$\tan^2\theta = \sec^2\theta - 1 = \frac{169}{25} - 1 = \frac{144}{25}$, and in QII $\tan\theta < 0$: $\tan\theta = -\frac{12}{5}$. Since $\cos\theta = \frac{1}{\sec\theta} = -\frac{5}{13}$: $\sin\theta = \tan\theta\cos\theta = \left(-\frac{12}{5}\right)\left(-\frac{5}{13}\right) = \frac{12}{13}$.

14 [Intermediate] • Target: Identities with squared expressions

Verify the identity $\frac{\csc x - 1}{\cot x} = \frac{\cot x}{\csc x + 1}$ by cross-multiplying and using the Pythagorean identity.

Cross-multiplying: $(\csc x - 1)(\csc x + 1) = \cot^2 x$, i.e. $\csc^2 x - 1 = \cot^2 x$, which is exactly the Pythagorean identity $1 + \cot^2 x = \csc^2 x$ rearranged — confirming the identity holds.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Identities with sums and products • Cross-Lesson: 8.01 Rational Functions and Asymptotes

Find all solutions to the trigonometric equation $2\sin^2 x - 3\sin x + 1 = 0$ on the domain $x \in [0, 2\pi)$.

Step 1: Factor the quadratic in $\sin x$: $(2\sin x - 1)(\sin x - 1) = 0$. Step 2: Solve each factor branch:Branch 1: $2\sin x - 1 = 0 \Rightarrow \sin x = \frac{1}{2}$. On $[0, 2\pi)$, $x = \frac{\pi}{6}$ (Quadrant I) or $x = \frac{5\pi}{6}$ (Quadrant II).Branch 2: $\sin x - 1 = 0 \Rightarrow \sin x = 1$. On $[0, 2\pi)$, $x = \frac{\pi}{2}$.The complete solution set is $x \in \left\{\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}\right\}$.

16 [Challenging] • Target: Identities involving all three Pythagorean forms • Cross-Lesson: 8.02 Graphing Rational Functions

[Unit Circle & Trigonometric Identities · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

17 [Challenging] • Target: Simplifying more complex expressions • Cross-Lesson: 8.03 Introduction to Limits

Rotate the Cartesian axes to eliminate the cross-product term in $5x^2 - 4xy + 8y^2 = 36$. Find the angle of rotation θ, standard form in the rotated system (x', y'), and the lengths of the semi-major and semi-minor axes.

Step 1: Determine the rotation angle θ: Here $A = 5$, $B = -4$, and $C = 8$.
$\cot(2\theta) = \frac{A - C}{B} = \frac{5 - 8}{-4} = \frac{-3}{-4} = \frac{3}{4} > 0$. Since $\cot(2\theta) = 3/4$, by right triangle trigonometry $\cos(2\theta) = \frac{3}{5}$.
Using half-angle formulas for $\theta \in (0, \pi/4)$: $\cos\theta = \sqrt{\frac{1 + \cos(2\theta)}{2}} = \sqrt{\frac{1 + 3/5}{2}} = \sqrt{\frac{4}{5}} = \frac{2}{\sqrt{5}}$, .

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18 [Challenging] • Target: Verifying identities with a common denominator strategy • Cross-Lesson: 9.01 Arithmetic Sequences and Series

Prove by mathematical induction that for all integers $n \geq 1$: $1 + 3 + 5 + \ldots + (2n - 1) = n^2$.

1. Base Step ($n = 1$): LHS $= 2(1) - 1 = 1$, RHS $= 1^2 = 1$. Since LHS = RHS, the base case $P(1)$ is true. 2. Inductive Hypothesis: Assume $P(k)$ is true for some integer $k \geq 1$: $1 + 3 + 5 + \ldots + (2k - 1) = k^2$. 3. Inductive Step ($n = k + 1$): Show that $1 + \ldots + (2k - 1) + (2(k + 1) - 1) = (k + 1)^2$: LHS $= [1 + 3 + \ldots + (2k - 1)] + (2k + 1)$. By the inductive hypothesis, substitute k^2.

19 [Challenging] • Target: Numeric identity verification (checking a specific angle) • Cross-Lesson: 9.02 Geometric Sequences and Series

Solve the logarithmic equation $\log_2(x + 3) + \log_2(x - 3) = 4$ for all real solutions x. State any extraneous roots.

Step 1: State domain restrictions: $x + 3 > 0 \Rightarrow x > -3$ and $x - 3 > 0 \Rightarrow x > 3$, so $x > 3$. Step 2: Combine logarithms: $\log_2[(x + 3)(x - 3)] = 4 \Rightarrow \log_2(x^2 - 9) = 4$. $x^2 - 9 = 2^4 = 16 \Rightarrow x^2 = 25 \Rightarrow x = \pm 5$. Step 3: Eliminate extraneous solutions: $x = -5$ violates $x > 3$ (producing negative arguments), so it is extraneous. The unique valid solution is $x = 5$.

20 [Challenging] • Target: Identities with squared expressions • Cross-Lesson: 9.03 The Binomial Theorem

An infinite geometric series has common ratio $r = -0.4$ and converges to an infinite sum of $S_\infty = 25$. Find the first term a_1, and compute the exact value of the partial sum S_4.

Use the infinite sum formula $S_\infty = \frac{a_1}{1 - r}$: $25 = \frac{a_1}{1 - (-0.4)} = \frac{a_1}{1.4} \Rightarrow a_1 = 25 \times 1.4 = 35$. Compute the 4th partial sum $S_4 = \frac{a_1(1 - r^4)}{1 - r}$: $(-0.4)^4 = 0.0256$. $S_4 = \frac{35(1 - 0.0256)}{1.4} = \frac{35(0.9744)}{1.4} = 25 \times 0.9744 = 24.36$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Using the Pythagorean identities

Divide $f(x) = 2x^3 + 5x^2 + 5x + 2$ by $(x + 1)$ using synthetic division. Find the quotient.

a. $2x^2 + 3x + 2$ ✓

b. $2x^2 + 3x - 2$

c. $2x^2 - 3x + 2$

d. $3x^2 + 3x + 2$

Synthetic division of $2x^3 + 5x^2 + 5x + 2$ by $x + 1$ gives quotient $2x^2 + 3x + 2$ with remainder 0.

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22 [Foundational] • Target: Simplifying and verifying

Consider the complex number $z = r(\cos\theta + i\sin\theta)$. If $z^3 = -8$ and $r > 0$, and the principal argument of z is $\frac{\pi}{3}$, find the value of r.

a. **2** ✓

b. 4

c. 8

d. 16

We are given the complex number $z = r(\cos\theta + i\sin\theta)$, where $r > 0$. We are also given that $z^3 = -8$ and the principal argument of z is $\theta = \frac{\pi}{3}$. Using De Moivre's Theorem, we have $z^3 = r^3(\cos(3\theta) + i\sin(3\theta))$. We are given $z^3 = -8$, which can be written in polar form as $8(\cos(\pi + 2k\pi) + i\sin(\pi + 2k\pi))$ for any integer k. Equating the magnitudes, we get $r^3 = 8$. Since $r > 0$, we have $r = \sqrt[3]{8} = 2$. Equating the arguments, we get.

23 [Foundational] • Target: Quotient and reciprocal identities

Consider the exponential growth model described by the function $P(t) = P_0e^{kt}$, where $P(t)$ is the population at time t, P_0 is the initial population, and k is the growth constant. If the initial population P_0 is 500 and the population after 4 years is 2000, determine the value of the growth constant k to three decimal places.

a. **0.347** ✓

b. 0.416

c. 0.301

d. 0.477

The population model is given by $P(t) = P_0e^{kt}$. We are given that $P_0 = 500$. We are also given that $P(4) = 2000$. Substituting these values into the equation, we get: $2000 = 500e^{k \cdot 4}$ Divide both sides by 500: $4 = e^{4k}$ Take the natural logarithm of both sides: $\ln(4) = \ln(e^{4k})$ $\ln(4) = 4k$ Solve for k: $k = \frac{\ln(4)}{4}$ Using a calculator, $\ln(4) \approx 1.38629$. $k \approx \frac{1.38629}{4}$ $k \approx 0.34657$ Rounding to three decimal places, $k \approx 0.347$.

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24 [Foundational] • Target: More identity verifications

A complex number z is represented in polar form as $z = r(\cos(\theta) + i\sin(\theta))$. If a complex number has a modulus $r = 5$ and an argument $\theta = \frac{\pi}{3}$, what is its rectangular form $a + bi$?

a. $\frac{5}{2} + i\frac{5\sqrt{3}}{2}$ ✓

b. $5 + i5$

c. $\frac{5\sqrt{3}}{2} + i\frac{5}{2}$

d. $5i$

The rectangular form of a complex number given in polar form $z = r(\cos(\theta) + i\sin(\theta))$ is $a + bi$, where $a = r\cos(\theta)$ and $b = r\sin(\theta)$. Given $r = 5$ and $\theta = \frac{\pi}{3}$: $a = 5\cos(\frac{\pi}{3}) = 5 \times \frac{1}{2} = \frac{5}{2}$
$b = 5\sin(\frac{\pi}{3}) = 5 \times \frac{\sqrt{3}}{2} = \frac{5\sqrt{3}}{2}$ Therefore, the rectangular form is $\frac{5}{2} + i\frac{5\sqrt{3}}{2}$.

25 [Foundational] • Target: Identities with sums and products

Find all real roots of $f(x) = x^3 - 4x^2 - 36x + 144$.

a. x=-6

b. x=-6,4

c. x=-5,4,6

d. **x=-6,4,6** ✓

Factoring: $f(x) = (x + 6)(x - 4)(x - 6) \Rightarrow x = -6, 4, 6$.

26 [Foundational] • Target: Identities involving all three Pythagorean forms

Find the common difference of the arithmetic sequence: -1, -2, -3, -4, ...

a. 0

b. -2

c. 1

d. **-1** ✓

Subtracting consecutive terms: $-2 - (-1) = -1$.

27 [Foundational] • Target: Simplifying more complex expressions

Find the 4th term of the geometric sequence with $a_1 = 2$ and common ratio $r = 2$.

a. 18

b. 17

c. 32

d. 16 ✓

$a_n = a_1 r^{n-1} = 2 \cdot (2)^3 = 16$.

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28 [Foundational] • Target: Verifying identities with a common denominator strategy

Let $(a_n)_{n \geq 1}$ be an arithmetic sequence of real numbers with first term $a_1 > 0$ and non-zero common difference d. The terms a_1, a_3, and a_4 form the first three terms of a convergent geometric sequence $(g_k)_{k \geq 1}$, such that $g_1 = a_1$, $g_2 = a_3$, and $g_3 = a_4$. Let $S_\infty = \sum_{k=1}^{\infty} g_k$ denote the sum to infinity of the geometric sequence, and let $S_n = \sum_{j=1}^{n} a_j$ denote the n-th partial sum of the arithmetic sequence. If $M = \max_{n \geq 1} S_n$ is the maximum value attained by the partial sums of the arithmetic sequence, what is the value of $\frac{M}{S_\infty}$?

a. $\frac{5}{4}$ ✓

b. $\frac{81}{64}$

c. $\frac{9}{8}$

d. $\frac{5}{8}$

1. **Determine the relation between d and a_1:** The terms of the arithmetic sequence are $a_n = a_1 + (n-1)d$. Therefore: $g_1 = a_1, \quad g_2 = a_3 = a_1 + 2d, \quad g_3 = a_4 = a_1 + 3d$ Since g_1, g_2, g_3 form a geometric progression, the geometric mean property requires: $g_2^2 = g_1 g_3 \Rightarrow (a_1 + 2d)^2 = a_1(a_1 + 3d)$ Expanding both sides: $a_1^2 + 4a_1 d + 4d^2 = a_1^2 + 3a_1 d \; a_1 d + 4d^2 = 0 \Rightarrow d(a_1 + 4d) = 0$ Since $de0$, we have $a_1 + 4d = 0$, which gives.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Numeric identity verification (checking a specific angle)

Using a double-angle identity, evaluate $\cos^2\theta - \sin^2\theta$ when $\theta = 15^\circ$.

a. -1

b. $-\frac{\sqrt{3}}{2}$

c. $\frac{\sqrt{2}}{2}$

d. $\frac{\sqrt{3}}{2}$ ✓

$\cos^2\theta - \sin^2\theta = \cos(2\theta)$; at $\theta = 15^\circ$, $2\theta = 30^\circ$, so the value is $\frac{\sqrt{3}}{2}$.

30 [Intermediate] • Target: Identities with squared expressions

Find the center and radius of $(x+6)^2 + (y-1)^2 = 25$.

a. Center $(6, -1)$, radius 5

b. **Center $(-6, 1)$, radius 5** ✓

c. Center $(1, -6)$, radius 5

d. Center $(-6, 1)$, radius 25

Standard form $(x-h)^2 + (y-k)^2 = r^2$ gives center $(h, k) = (-6, 1)$ and $r^2 = 25$, so $r = 5$.

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31 [Intermediate] • Target: Using the Pythagorean identities

Using complex numbers, find $\cos^4\theta$ as a sum of cosines.

a. $\frac{3+4\cos 2\theta+\cos 4\theta}{8}$ ✓

b. $\frac{1+\cos 4\theta}{2}$

c. 3/8

d. $\cos 4\theta/8$

$\cos^4\theta = \left(\frac{e^{i\theta}+e^{-i\theta}}{2}\right)^4$. Expand binomially.

32 [Intermediate] • Target: Simplifying and verifying

What is the center of the circle $(x-3)^2+(y+2)^2=16$?

a. (-3, 2)

b. **(3, -2)** ✓

c. (3, 2)

d. (-2, 3)

Comparing to $(x-h)^2+(y-k)^2=r^2$: center $(h,k)=(3,\ -2)$.

33 [Intermediate] • Target: Quotient and reciprocal identities

For $z=re^{i\theta}$, express $\mathrm{Re}(z^3)$ using triple angle formula.

a. r^3

b. $3r^3\cos\theta$

c. $r^3\cos 3\theta$ ✓

d. $r\cos 3\theta$

$z^3=r^3e^{3i\theta}$. $\mathrm{Re}(z^3)=r^3\cos 3\theta$.

34 [Intermediate] • Target: More identity verifications

Evaluate $\sum_{k=1}^{n}\cos k\theta$ using complex exponentials.

a. 0

b. $n\cos\theta$

c. $\frac{\sin(n\theta/2)}{\sin(\theta/2)}\cos\frac{(n+1)\theta}{2}$ ✓

d. $\cos(n\theta)$

Geometric series of $e^{ik\theta}$, take real part.

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35 [Intermediate] • Target: Identities with sums and products

State $\cos 2\theta$ in three equivalent forms.

a. $\cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta$ ✓

b. $\cos\theta + \sin\theta$

c. $2\sin\theta\cos\theta$

d. $\cos^2\theta$

Three forms of double angle for cosine.

36 [Intermediate] • Target: Identities involving all three Pythagorean forms

A sequence is defined by the recurrence relation $a_{n+1} = a_n^2 - 2$ for $n \geq 1$. If $a_1 = 2$, find the value of a_{2023}.

a. **2** ✓

b. 0

c. -2

d. 1

Let's compute the first few terms of the sequence: $a_1 = 2$ $a_2 = a_1^2 - 2 = 2^2 - 2 = 4 - 2 = 2$ $a_3 = a_2^2 - 2 = 2^2 - 2 = 4 - 2 = 2$ It appears that for $a_1 = 2$, all subsequent terms of the sequence are also 2. We can prove this by induction. Base case: For $n = 1$, $a_1 = 2$, which is true. Inductive step: Assume that for some integer $k \geq 1$, $a_k = 2$. We need to show that $a_{k+1} = 2$. From the recurrence relation, $a_{k+1} = a_k^2 - 2$. By the inductive hypothesis, $a_k = 2$. Substituting this into.

37 [Intermediate] • Target: Simplifying more complex expressions

A population of bacteria, initially P_0 cells, grows according to the logistic model $P(t) = \frac{L}{1 + Ae^{-kt}}$, where t is in hours, L is the carrying capacity, and A and k are positive constants. At $t = 0$, the population is $P(0) = \frac{L}{3}$. After 2 hours, the population has doubled to $P(2) = \frac{2L}{3}$. Find the exact value of e^{2k} in terms of L and P_0.

a. $\frac{9}{4}$ ✓

b. $\frac{4}{9}$

c. 3

d. $\frac{3}{2}$

The logistic model is given by $P(t) = \frac{L}{1 + Ae^{-kt}}$. At $t = 0$, $P(0) = \frac{L}{3}$. Substituting into the model: $\frac{L}{3} = \frac{L}{1 + Ae^{-k \cdot 0}}$ $\frac{L}{3} = \frac{L}{1 + A}$ $1 + A = 3$ $A = 2$. So the model becomes $P(t) = \frac{L}{1 + 2e^{-kt}}$. We are also given that $P(2) = \frac{2L}{3}$. Substituting into the model: $\frac{2L}{3} = \frac{L}{1 + 2e^{-2k}}$ $\frac{2}{3} = \frac{1}{1 + 2e^{-2k}}$ $2(1 + 2e^{-2k}) = 3$ $2 + 4e^{-2k} = 3$ $4e^{-2k} = 1$ $e^{-2k} = \frac{1}{4}$. We need.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Verifying identities with a common denominator strategy • Cross-Lesson: 9.01 Arithmetic Sequences and Series

Consider the exponential function $f(x) = 2^x$. If $f(a) = 8$, what is the value of a?

a. 1

b. 2

c. **3** ✓

d. 4

The given function is $f(x) = 2^x$. We are given that $f(a) = 8$. Substituting a into the function, we get $2^a = 8$. To solve for a, we need to express 8 as a power of 2. Since $2^3 = 2 \times 2 \times 2 = 8$, we can rewrite the equation as $2^a = 2^3$. Since the bases are equal, the exponents must be equal. Therefore, $a = 3$.

39 [Challenging] • Target: Numeric identity verification (checking a specific angle) • Cross-Lesson: 9.02 Geometric Sequences and Series

Consider the exponential function $f(x) = 2^x$. What is the value of $f(3)$?

a. 6

b. **8** ✓

c. 9

d. 5

The function is given by $f(x) = 2^x$. To find $f(3)$, we substitute $x = 3$ into the function: $f(3) = 2^3$. Calculating the power, $2^3 = 2 \times 2 \times 2 = 8$. Therefore, the value of $f(3)$ is 8.

40 [Challenging] • Target: Identities with squared expressions • Cross-Lesson: 9.03 The Binomial Theorem

Find the vector projection of $\mathbf{u} = \langle 3, 4 \rangle$ onto $\mathbf{v} = \langle 2, 0 \rangle$.

a. $\langle 0, 3 \rangle$

b. $\langle 4, 0 \rangle$

c. $\langle 3, 0 \rangle$ ✓

d. $\langle 3, 4 \rangle$

$\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}}\mathbf{v} = \frac{3}{2}\langle 2, 0 \rangle = \langle 3, 0 \rangle$.

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41 [Challenging] • Target: Using the Pythagorean identities • Cross-Lesson: 1.01 Functions and Function Notation

For $f(x) = \frac{x^2 - 25}{x - 5}$ when $x \neq 5$, and $f(5) = k$, find k so that f is continuous at $x = 5$.

a. **10** ✓

b. 5

c. 9

d. 11

$\lim_{x \to 5} f(x) = \lim_{x \to 5} (x + 5) = 10$ (after canceling $x - 5$). For continuity, k must equal this limit: $k = 10$.

42 [Challenging] • Target: Simplifying and verifying • Cross-Lesson: 1.02 Domain and Range

The arithmetic sequence with $a_1 = 3$ and $d = 4$ has $a_n = 99$. Find n.

a. **25** ✓

b. 99

c. 26

d. 24

Applying Sequences principles, $3 + (n - 1)(4) = 99 \Rightarrow n = 25$. Therefore, the verified correct option is 25.

43 [Challenging] • Target: Quotient and reciprocal identities • Cross-Lesson: 1.03 Average Rate of Change

Solve for the larger value of x: $2^{2x} - 10 \cdot 2^x + 16 = 0$.

a. 5

b. **3** ✓

c. 1

d. 4

Let $u = 2^x$: $u^2 - 10u + 16 = 0 \Rightarrow (u - 2^1)(u - 2^3) = 0 \Rightarrow x = 1$ or 3. Larger: $x = 3$.

44 [Challenging] • Target: More identity verifications • Cross-Lesson: 2.01 Right-Triangle Trigonometry

For the parabola $y^2 = -20x$, find the vertex, focus, and directrix.

a. **Vertex** $(0, 0)$**, focus** $(-5, 0)$**, directrix** $x = 5$ ✓

b. Vertex $(0, 0)$, focus $(-10, 0)$, directrix $x = 10$

c. Vertex $(0, 0)$, focus $(0, -5)$, directrix $y = 5$

d. Vertex $(-5, 0)$, focus $(0, 0)$, directrix $x = -5$

Comparing $y^2 = 4px$ to $y^2 = -20x$: $4p = -20$, so $p = -5$. Vertex $(0, 0)$, focus $(-5, 0)$, directrix $x = 5$.

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45 [Challenging] • Target: Identities with sums and products • Cross-Lesson: 2.02 Radian Measure and Arc Length

Find the 10th term of the arithmetic sequence with $a_1 = 3$ and $d = 4$.

a. 36

b. 35

c. **39** ✓

d. 43

Applying Sequences principles, $a_{10} = 3 + (10 - 1)(4) = 39$. Therefore, the verified correct option is 39.

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